

Xinxian Ma

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Education

Ecole Polytechnique Fédérale de Lausanne (EPFL), MS in Data Science Sept 2025 – Present

- **GPA:** 5.75/6.0
- **Course:** Machine Learning, Modern nature language processing, Applied Data Analysis

The Chinese University of Hong Kong (Shenzhen), BS in Data Science Sept 2021 – May 2025

- **GPA:** 3.9/4.0 (Top 2%)
- **Course:** Machine Learning, Deep Learning, Advanced Machine Learning

Research Experience

Influence-guided evolutionary search for post-training data Sep 2026 – Present
Research Assistant, Computer Vision Lab (CVLab), EPFL
Supervisor: Prof. Mathieu Salzmann

- Develop influence-guided data optimization methods that iteratively search, generate, and refine high-value training data for more efficient model improvement.
- Build large-scale experimentation and evaluation pipelines for data-centric LLM training, studying how trainin

zip2zip-Based Compression-Aware Language Modeling Feb 2026 – Present
Research Assistant, Data Science & AI Lab (DLAB), EPFL
Supervisor: Prof. Robert West
Manuscript under review at ICLR 2027

- Develop and train compression-aware LLMs with inference-time adaptive tokenization, enabling 15–40% fewer input/output tokens while preserving competitive language modeling and downstream performance.
- Diagnose failure modes of compressed-token modeling and develop architectural and training improvements, including decoupled input/output representations, improved initialization, and span-aware RoPE positioning.
- Build efficient generation and evaluation infrastructure, including tree-attention-based batched decoding and multi-view evaluation, and speculative decoding; support end-to-end pretraining/SFT, LoRA and full fine-tuning on LLaMA- and Phi-based models.

Federated Data Visualization and Clustering Apr 2024 – Aug 2024
Research Assistant, Shenzhen Research Institute of Big Data
Supervisor: Prof. Jicong Fan
Shenzhen, China

- Proposed Fed-tSNE and Fed-UMAP for privacy-preserving high-dimensional data visualization in federated learning, leveraging kernel methods and maximum mean discrepancy.
- Extended the framework to distributed clustering algorithms, including spectral clustering, and conducted large-scale experiments to evaluate effectiveness and scalability.

Publications

Federated t-SNE and UMAP for Distributed Data Visualization (**AAAI 2025**) Sept 2024
Dong Qiao, **Xinxian Ma**, Jicong Fan [arXiv:2412.13495](#)

Internship Experience

Data Scientist Intern, Flywheel – Shenzhen, China Jun 2025 – Aug 2025

- Built standardized sales and media performance analytics pipelines using Amazon Marketing Cloud (AMC) SQL, enabling audience package creation and multi-dimensional cross-campaign benchmarking
- Produced custom insights and activation strategies for key brands by analyzing incremental customer acquisition, lifecycle value, and purchase paths to optimize targeting and media spend.

Data Scientist Intern, Anker – Shenzhen, China

Feb 2025 – Aug 2025

- Performed end-to-end EDA on sales data, including anomaly detection and treatment to improve data reliability for downstream modeling.
- Built demand forecasting pipelines using time-series models and LightGBM to support inventory planning.
- Developed product profile via clustering using wavelet denoising and DTW-based time-series K-means

AI Engineer Intern, Fandao – Guangzhou, China

Jun 2024 – Aug 2024

- Contributed to customized AI content generation by participating in fine-tuning, model testing, dataset preparation, and facilitated workflow design to ensure robust business solution implementation
- Assisted in development of an AI customer service system using Retrieval-Augmented Generation (RAG), enhancing product sales and department satisfaction

Projects

RPNR: Indirect Prompt Injection Benchmark for GitHub MCP Agents

[code](#)

- Built an indirect prompt injection benchmark for GitHub MCP coding agents across repository files, issues, pull requests, and comments, with an end-to-end evaluation framework for multiple agents and defenses.
- Evaluated 384 tasks across 320 GitHub repositories, achieving >90% clean-task success while observing attack success rates of up to 14.9%.

Smiley Prediction in Twitter Sentiment Data

[code](#)

- Built a tweet sentiment classification pipeline using GloVe/FastText baselines and fine-tuned TweetNLP and BERTweet models.
- Achieved 92.0% on the public leaderboard (Rank #3).

N-square queens problem with MCMC

[code](#)

- Implements Markov Chain Monte Carlo methods for the 3D (N^2)-Queens problem using the Metropolis–Hastings algorithm with simulated annealing, resulting in a stable and convergent solution approach (full score).

SwissReach: Interactive Visualization of Swiss Public Transport Accessibility

[code](#)

- Built an interactive data visualization platform using Swiss public transport timetable data, combining graph-based reachability analysis to explore nationwide mobility and access to everyday services (full score).

Skill

Technical: Python, C++, C, Java, R, Matlab, SQL, JavaScript

Language: English (IELTS 7.5), Mandarin (Native)

Award

Mathematical Contest in Modeling (Top 5%)

Feb 2024

Dean's List (2021, 2022, 2023, 2024)

2021 – 2024

CUHK(sz) Academic Scholarship

2021 - 2024

Others

Teaching Assistant, EPFL CS-401 Applied Data Analysis

[2026 Fall](#)

Teaching Assistant, CUHK(sz) STA2003 Statistics & Probability

[2023 Fall](#)

Aid Education Volunteer Certificate Of Appreciation

soapcycling.org

Hobby

Basketball, Football, Hiking, Drawing, Photography